

Project ID :

R25-036

1. Topic (12 words max)

Generative AI-Based Chatbot for Employee and Customer Support Automation in LOLC Bank

2. Research group the project belongs to

CEAI - Centre of Excellence in AI

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	-
Year	-

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Organizations today face challenges in managing employee queries across various departments, such as HR and IT, often involving repetitive questions about policies, workflows, and technical support. For LOLC, the current system [1] handles these operations but lacks a centralized solution to address employee queries efficiently. This creates communication bottlenecks, consumes time, increases operational costs, and impacts overall productivity.

This research focuses on designing a chatbot that integrates seamlessly with LOLC's internal system to streamline communication, automate routine tasks, and enhance the employee experience. Acting as a centralized knowledge base, the chatbot will provide instant answers to queries, reducing time spent on repetitive tasks, lowering operational costs, and enabling employees to focus on more critical responsibilities. Key challenges include ensuring compatibility with existing systems, minimal disruption during implementation, and scalability.

Deployment will follow a phased approach, starting with a pilot in a smaller department to gather feedback and refine functionality. Once validated, the chatbot will be scaled organization-wide using structured deployment and robust user training. Integration with HR and IT systems will utilize API gateways for smooth data flow, while feedback loops will enable continuous improvement.

Performance monitoring will ensure long-term success, with metrics such as response time, query resolution rate, user satisfaction, and cost savings tracked in real-time. The chatbot's NLP engine[2] [3] will adapt over time to enhance query understanding and accuracy. Regular evaluations will ensure the system remains efficient and relevant, contributing to reduced operational expenses by automating routine tasks and streamlining processes [4].

The ultimate goal is to create an intelligent, adaptive chatbot that reduces response times, improves employee satisfaction, lowers operational costs, and drives organizational efficiency through seamless integration and continuous learning [5].

References:

[1].lolc.oasys.lk

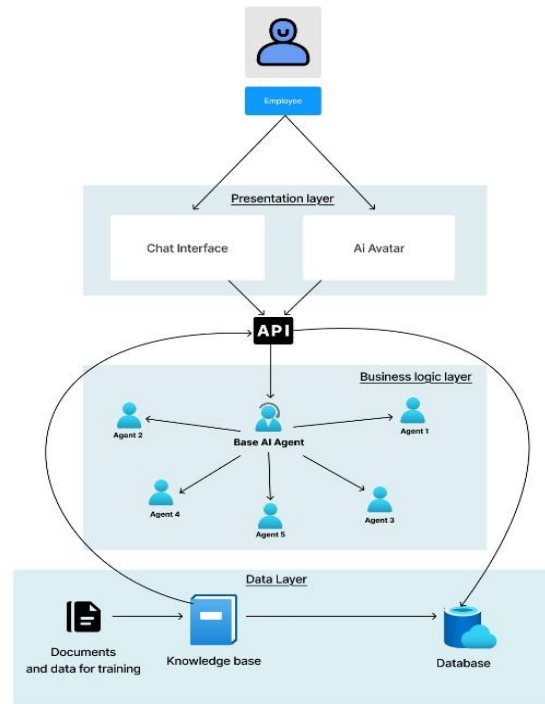
[2].<https://www.deeplearning.ai/>

[3] <https://huggingface.co/>

[4].https://www.researchgate.net/publication/378149233_History_of_generative_Artificial_Intelligence_AI_chatbots_past_present_and_future_development

[5].https://www.researchgate.net/publication/342842655_The_Impact_of_Chatbots_on_Customer_Service_Performance

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)



We are developing an advanced AI-powered chatbot solution designed to streamline internal communication and enhance the employee experience at LOLC. The chatbot combines a dynamic knowledge-based system with cutting-edge features, including voice-to-voice support, 3D avatar lip-syncing, AI-driven avatar generation, and multi-agent functionality. This innovative system provides prompt and efficient guidance to employees, delivering instant answers to HR policy inquiries, IT support requests, and other business process-related questions.

The solution integrates a robust knowledge base that updates dynamically to ensure accurate and relevant responses. Its voice support feature facilitates seamless interactions, catering to a diverse workforce. The AI avatar generation adds a personalized and engaging dimension to the experience, while multi-agent capabilities enable the chatbot to handle multiple simultaneous queries effectively. Additionally, it integrates with forms automation workflows, leveraging knowledge base responses to streamline processes.

By simplifying access to critical information and providing timely solutions, the chatbot significantly reduces response times, minimizes repetitive workloads for HR and IT teams, and enhances overall employee productivity.

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

1. AI and Machine Learning (ML):

Expertise in natural language processing (NLP) and ML is essential for building a chatbot that understands diverse queries, identifies user intent, and generates appropriate responses. Knowledge in AI-driven voice recognition and avatar generation is also critical for providing an engaging and personalized user experience.

2. Knowledge Base Development:

A comprehensive understanding of LOLC's HR policies, IT processes, and internal business operations is necessary to create an accurate and dynamic knowledge base. Collaboration with domain experts ensures that the knowledge base reflects the most up-to-date and relevant information.

3. Multi-Agent Systems:

Experience in designing and managing multi-agent architectures is required to enable the chatbot to handle concurrent interactions seamlessly. This includes load balancing and session management to maintain performance during peak usage.

4. User Experience (UX) Design:

Creating an intuitive, accessible, and engaging chatbot interface involves UX design expertise. This ensures that the chatbot aligns with LOLC's branding and is user-friendly for employees with varying technical proficiency levels.

Data Requirements:

1. Internal HR and IT Documentation: Accurate and updated HR policies, IT support workflows, and operational guidelines form the foundation of the knowledge base.

2. IT Support Data: IT troubleshooting guides and FAQs, System architecture documentation for integration & Historical IT tickets and resolutions for AI training.

3. Employee Feedback Data: Surveys and feedback forms about common queries and Suggestions for improving existing processes or systems.

4. Integration Data: API endpoints for LOLC's internal systems, Internal data schemas and data flow mappings & User authentication and authorization protocols for secure access.

5. Natural Language Processing (NLP) Training Data: Textual datasets with domain-specific terminologies and Real-world query examples from employees (anonymized), Synonyms, abbreviations, and contextual phrases unique to LOLC.

8. Objectives and Novelty
Main Objective:

The primary objective of this project is to develop an intelligent chatbot tailored to LOLC's needs, designed to enhance employee communication and streamline operational processes. This chatbot will serve as a centralized platform for addressing common queries related to HR policies, IT support, and other business operations, providing employees with instant and accurate responses.

The solution aims to improve efficiency across departments while allowing employees to focus on more critical responsibilities. Additionally, the chatbot will integrate seamlessly with LOLC's existing systems, ensuring a smooth adoption process and scalability for future organizational needs.

Member Name	Sub Objective	Tasks	Novelty
Fernando W.S.N.	Design and implement the chatbot's digital avatar to enhance user engagement and interaction.	- Create a visually appealing and responsive avatar that aligns with LOLC branding. - Implement interactive features like animations and expressions. - Implement voice-to-voice interaction with the avatar.	Introduces a visually engaging and dynamic element to the chatbot, making it more interactive and user-friendly.
Gamage U. R	Develop and maintain a centralized knowledge base, train it with relevant LOLC's HR and IT support data, and ensure it delivers accurate responses based on employee queries.	- Domain-Specific Model Creation for HR and IT Support - Build and maintain an up-to-date, searchable knowledge base.	Provides a robust and reliable knowledge base for HR policies and IT support, ensuring it processes and delivers the necessary results to the backend.

		- Implement Retrieval-Augmented Generation (RAG) System	
Lakshan S.N.	Implement intelligent processes that enable AI agents to analyze and deliver relevant responses to employee queries based on the knowledge base and context.	- Implement Natural Language Processing (NLP) for query understanding. - Implement Multi-Agent System for Collaborative Task Automation - Implement the backend API's	Leverages AI to create a contextually aware chatbot capable of handling diverse and complex queries.
Kuruppu K.A.V. U	Facilitate seamless integration of the chatbot with LOLC's internal forms and systems.	- Analyze current forms and internal data structures. - Implement features for data retrieval and autofill in forms. - Ensure compatibility and smooth data flow between systems. - Add Auto-Suggestions and Highlights for Improved User Experience. - User Profile Creation and Chat History Storing	Enables automation of form handling processes, reducing manual effort and errors. The chatbot can use previous data to preemptively address similar problems.

9. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Prof.	Nuwan	Kodagoda	<i>Nuwan Kodagoda</i> (signed)  Confirmation.eml
Co-Supervisor	Dr.	Lakmini	Abeywardhana	DLABeywardhana (Signed)  Confirmation.eml
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	
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- b) Does the proposed topic exhibit novelty?

Yes		No	
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

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Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.